Problem 3 | Problems T3_Diffusion.pdf #3 | Two-Point Arrhenius · Activation Energy

Activation Energy and Diffusivity of Carbon in Mild Steel

Original statement · Problems T3_Diffusion.pdf

3.- The diffusivity of carbon in mild steel was measured at two different temperatures:
Data: $R = 8.314\ \mathrm{J/mol \cdot K}$
• *At* $T_1 = 850^\circ\mathrm{C}$*:* $D_1 = 4.826 \times 10^{-12}\ \mathrm{m^2/s}$
• *At* $T_2 = 950^\circ\mathrm{C}$*:* $D_2 = 1.805 \times 10^{-11}\ \mathrm{m^2/s}$
Using these data, calculate:
a) The activation energy for the C *diffusion in mild steel, for the temperature interval of* $850^\circ\mathrm{C}$ *to* $950^\circ\mathrm{C}$ *(Solution:* $E_D = 36\ \mathrm{kcal/mol}$*)*
b) The diffusivity at $1000^\circ\mathrm{C}$*, assuming that the activation energy does not change at this temperature range. (Solution:* $D = 3.23 \times 10^{-11}\ \mathrm{m^2/s}$*)*

Official Solution: ED = 36 kcal/mol; D(1000 °C) = 3.23 x 10^-11 m^2/s

PHASE 1 · HYPOTHESES & PARAMETERS

- **Single Arrhenius Law (assumption):** The problem uses one Arrhenius law with constant D_0 and E_D for C in mild steel from $850^\circ\mathrm{C}$ to $1000^\circ\mathrm{C}$, as part b) states [Session 5 Slide 26].
- **Parameters:** $T_1 = 850^\circ\mathrm{C} = 1123.15\ \mathrm{K}$, $D_1 = 4.826 \times 10^{-12}\ \mathrm{m^2/s}$ $T_2 = 950^\circ\mathrm{C} = 1223.15\ \mathrm{K}$, $D_2 = 1.805 \times 10^{-11}\ \mathrm{m^2/s}$ $T_3 = 1000^\circ\mathrm{C} = 1273.15\ \mathrm{K}$, $R = 8.314\ \mathrm{J/mol \cdot K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

Dividing Arrhenius equations at two temperatures eliminates D_0 [Session 5 Slide 26]:

$$\ln\left(\frac{D_2}{D_1}\right) = \frac{E_D}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \implies E_D = \frac{R \cdot \ln(D_2/D_1)}{\frac{1}{T_1} - \frac{1}{T_2}}$$

For part (b), extrapolating to $T_3 = 1000^\circ\mathrm{C}$:

$$D_3 = D_2 \cdot \exp\left[\frac{E_D}{R}\left(\frac{1}{T_2} - \frac{1}{T_3}\right)\right]$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

Part (a): Activation Energy E_D

$$\frac{D_2}{D_1} = \frac{1.805 \times 10^{-11}}{4.826 \times 10^{-12}} = 3.7401575 \implies \ln(3.7401575) \approx 1.319128$$

$$\frac{1}{T_1} - \frac{1}{T_2} = \frac{1}{1123.15} - \frac{1}{1223.15} = 8.903530 \times 10^{-4} - 8.175612 \times 10^{-4} = 7.27918 \times 10^{-5}\ \mathrm{K^{-1}}$$

$$E_D = \frac{8.314 \times 1.319128}{7.27918 \times 10^{-5}} = 150\,666\ \mathrm{J/mol} = 150.67\ \mathrm{kJ/mol}$$

$$E_D = \frac{150\,666\ \mathrm{J/mol}}{4184\ \mathrm{J/kcal}} \approx \mathbf{36.01}\ \mathrm{kcal/mol} \approx \mathbf{36}\ \mathrm{kcal/mol}$$

Part (b): Diffusivity at $1000^\circ\mathrm{C}$ ($T_3 = 1273.15$ K)

$$\frac{1}{T_2} - \frac{1}{T_3} = \frac{1}{1223.15} - \frac{1}{1273.15} = 8.175612 \times 10^{-4} - 7.854534 \times 10^{-4} = 3.21078 \times 10^{-5}\ \mathrm{K^{-1}}$$

$$\text{Exponent} = \frac{150\,666}{8.314} \times (3.21078 \times 10^{-5}) = 18121.96 \times 3.21078 \times 10^{-5} \approx 0.581857$$

$$D_3 = (1.805 \times 10^{-11}\ \mathrm{m^2/s}) \times e^{0.581857} = (1.805 \times 10^{-11}) \times 1.78936 \approx \mathbf{3.23 \times 10^{-11}}\ \mathrm{m^2/s}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Consistency with Official Data: The official reference table [Session 5 Slide 23] lists $D(\text{C in Fe-FCC}) \approx 3 \times 10^{-11}\ \text{m}^2/\text{s}$ at 1000°C, matching our derived $3.23 \times 10^{-11}\ \text{m}^2/\text{s}$; both are of the same order, the slide value being a one-significant-figure figure ($\approx 8\%$ lower). **Mechanism Check:** $E_D \approx 36\ \text{kcal/mol}$ is lower than the activation energy for self-diffusion of iron ($57.5\ \text{kcal/mol} = 240\ \text{kJ/mol}$ for α-Fe) [Slide 9]. This agrees with carbon diffusing interstitially, which needs less energy than vacancy diffusion [Slide 13].